VARUN ENGINEERS
HEAD OFFICE : # A-28/1, 3&4, Road No. 15, Behind GVK Bio Science,
IDA, Nacharam, Hyderabad - 500 076.
Ph: 72070 57694, 98665 01955, 98665 07955
E-mail : varunengineers@gmail.com, info@varunengineersindia.com

Quotation No : Q25000050

Date : 11-Feb-2025

Hetero Labs Ltd U-1
Sy.10, IDA, Gaddapotharam, Jinnaram, Sangareddy
HYDERABAD
TELANGANA
502319

Kind Attention : Mr./Ms. Mallikarjun

Dear Sir,

Our team understands that selecting gas generation systems is a difficult task.
We have the expertise to help you being a prime manufacturer and supplier of gas generation systems.

Absstem has the resources, facilities, technology, and expertise required to ensure a successful project.
Our approach is low-risk because of our previous experience with large and complex projects.
Post-installation, our shield service division, with diligent accuracy and functionality, oversees a team of technicians who provide warranty and post-warranty support to your newly-installed gas generation systems.
Our customers have relied on scale up their production.

Our products have consistently delivered absolute purity requirements.
The industry's lowest ever production downtime.
Reduce OPEX by at least 70%.

We wish to form a long-term relationship with the goal to develop the most cost-effective and comprehensive solutions for your needs.
With an experienced team of professional personnel already in place, our team is prepared to move swiftly to ensure that we meet critical objectives & timelines.
Please feel free to contact in case of any query.

Looking forward for a positive response.

Thanks and Regards,

For VARUN ENGINEERS

K.Ramesh

Managing Director

9866507955





nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN SH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

Absstem Industrial Solutions Pvt. Ltd.
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

Product Description

a. Product Name	High Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN SH**
c. Purity Range	99% - 99.95%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generator
- Nitrogen Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA system is stored in this tank for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

1) Guaranteed assurance of plant performance.
2) Genuine spare parts and consumables being used to provide longevity and durability.
3) Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
4) 24 x 7 On-Call service support to our customers for any basic trouble shooting
5) Emergency visit to attend any unplanned breakdown to any location within 48 hours.
6) No additional surprise charges as all breakdown spares and consumables covered.
7) The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

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Product Description

Product	Description	Make	Qty
Compressed Air System	C200F1330 Air System		1

Caption	Specification
Air Compressor KW	110+90 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	11000 Liters Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	1332 CFM
Air Pressure	7.5 Barg
POWER SUPPLY	400 V 3 Phase 50 Hz

Product	Description	Make	Qty
nGen Nitrogen Generator High Purity	nGEN160SH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99% PURITY	1095 Nm3/Hr DRY AIR REQUIRED 1418 CFM
FLOW AT 99.5% PURITY	916 Nm3/Hr DRY AIR REQUIRED 1375 CFM
FLOW AT 99.9% PURITY	680 Nm3/Hr DRY AIR REQUIRED 1289 CFM
FLOW AT 99.95% PURITY	612 Nm3/Hr DRY AIR REQUIRED 1351 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	12000 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	12000 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*



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Product	Description	Make	Qty
Gas Storage Vessel	AI-15000L Storage Vessel		1

Caption	Specification
Water Volume	15000 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	97.5 M3
Corrosion Allowance	1.5 MM

Product	Description	Make	Qty
4 x 5 Booster 300 Nm3/hr, 15 Barg	4 x 5 Booster 300 Nm3/hr, 15 Barg Booster		2

Caption	Specification
Suction Pressure	5 - 6 Barg
Discharge Pressure	14 - 15 Barg
Motor Power	15 KW
Shaft Power	11.28 KW
Flow Capacity	300 Nm3/hr
Type	water cooled non lubricated
Cooling Water Required	2.0 m3/hr



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Product	Description	Make	Qty
HP Storage Tank 20 Barg	5000 ltr HP Storage Tank 20 Barg High Pressure Storage Tank	ABSSTEM	2

Caption	Specification
Capacity	5000 litres
Working Pressure	20 Barg
Hydrotest Pressure	33 Barg
Shell / Dish Thickness	18 mm / 20 mm
In / Out Connection	80 NB
Design Standard	IS 2825

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	ACSC200F	C200F1330	Compressed Air System	840510	1	75,65,000.00	18.00	75,65,000.00
2	nGEN160SH	nGEN160SH	nGen Nitrogen Generator High Purity	840510	1	1,76,16,000.00	18.00	1,76,16,000.00
3	AIS1508ST	AI-15000L	Gas Storage Vessel	731100	1	10,00,000.00	18.00	10,00,000.00
4	BOS31	4 x 5 Booster 300 Nm3/hr, 15 Barg	4 x 5 Booster 300 Nm3/hr, 15 Barg	8414	2	26,30,000.00	18.00	52,60,000.00
5	HPS7	5000 ltr HP Storage Tank 20 Barg	HP Storage Tank 20 Barg	73110090	2	6,21,500.00	18.00	12,43,000.00

Sub Total	3,26,84,000.00
GST(%) @18.00	58,83,120.00
Grand Total	3,85,67,120.00



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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For VARUN ENGINEERS

K.Ramesh

Managing Director

9866507955

